

Shivvrat Arya

Newark NJ 07102

✉ shivvrat.arya@njit.edu | 🌐 shivvrat.github.io/ | 📄 Shivvrat | 📱 shivvrat | 📧 Shivvrat Arya

Professional Summary

Assistant Professor of Computer Science at the Ying Wu College of Computing, New Jersey Institute of Technology (NJIT). My research advances trustworthy, interpretable, and efficient artificial intelligence through methods that integrate neural learning, symbolic and probabilistic reasoning, and optimization. I develop algorithms for reliable decision-making, scalable inference, and structured reasoning, with applications in computer vision, video understanding, multimodal learning, and human-centered AI. My work spans neurosymbolic AI, tractable and approximate probabilistic inference, neural combinatorial optimization, explainable AI, graph-structured decision-making, and foundation models for reasoning across language and vision. My research has appeared at NeurIPS, AAAI, AISTATS, CIKM, ACM TiiS, and related venues, and has been recognized with a NeurIPS Spotlight, AAAI Oral Presentation, TPM Best Paper Award, and support from DARPA, NSF, and AFOSR-funded projects.

Education

Ph.D. in Computer Science

May 2025

THE UNIVERSITY OF TEXAS AT DALLAS, RICHARDSON, TEXAS

GPA: 4.0/4.0

- **Dissertation Title:** Neural Solvers for Fast, Accurate Probabilistic Inference
- **Advisor:** Dr. Vibhav Gogate
- **Co-Advisor:** Dr. Yu Xiang

M.S. in Computer Science

May 2021

THE UNIVERSITY OF TEXAS AT DALLAS, RICHARDSON, TEXAS

GPA: 4.0/4.0

B. Tech. in Computer Science and Engineering

May 2019

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY VADODARA, INDIA

GPA: 8.39/10.0

Research Areas

Artificial Intelligence (AI); Machine Learning (ML); Deep Learning (DL); Neuro-Symbolic and Trustworthy AI; Neural Combinatorial Optimization; Deep Reinforcement Learning (DRL) for Optimization and Control; Tractable and Approximate Probabilistic Inference; Scalable and Adaptive Decision-Making; Explainable and Interpretable AI (XAI/IAI); Generative and Foundation Models for Reasoning; Computer Vision and Video Understanding.

Academic Appointments and Research Experience

Department of Computer Science, Ying Wu College of Computing, NJIT

Newark, NJ

ASSISTANT PROFESSOR

September 2025 – Present

- Establishing an independent research program in trustworthy and efficient artificial intelligence, with emphasis on neurosymbolic AI, tractable inference, optimization, and foundation models for reasoning.
- Developing and teaching undergraduate honors and graduate courses in artificial intelligence and machine learning, including new offerings that strengthen the department's AI curriculum.
- Advising and mentoring students through research supervision and committee service, contributing to departmental curriculum and research service, and building research collaborations.

Center for Machine Learning, The University of Texas at Dallas

Richardson, TX

RESEARCH ASSISTANT

August 2021 - May 2025

- Developed neural and probabilistic inference methods for tractable models, probabilistic graphical models, and neurosymbolic systems in collaboration with faculty and research teams.
- Published work in leading AI/ML venues including NeurIPS, AAAI, AISTATS, and ACM TiiS, with research recognized through spotlight, oral presentation, and best paper honors.
- Contributed as a Graduate Research Assistant / research contributor to externally funded research programs supported by DARPA, NSF, and AFOSR.

Department of Computer Science, The University of Texas at Dallas

Richardson, TX

TEACHING ASSISTANT

August 2020 - August 2021

- **Courses:** Statistical Methods in AI and Machine Learning, Database Systems, Discrete Mathematics for Computing II
- **Responsibilities:** Graded assignments, held office hours, conducted project demonstrations, and developed supplementary materials and homework solutions for the courses.

Externally Funded Research Contributions

DARPA Perceptually-enabled Task Guidance (PTG) program DARPA

GRADUATE RESEARCH ASSISTANT

Aug 2021 - May 2025

- Advanced neurosymbolic dynamic models combining structured reasoning with deep learning.
- Improved user performance by expanding skillsets and reducing error rates in task guidance settings.
- Built robust pipelines for perception, inference, and feedback loops.

DARPA Explainable Artificial Intelligence (XAI) program DARPA

GRADUATE RESEARCH ASSISTANT

Aug 2020 - Aug 2021

- Delivered high-performance explainable models without sacrificing accuracy.
- Designed methods to increase transparency and user trust for decision support.

DARPA Assured Neuro Symbolic Learning and Reasoning (ANSR) program DARPA

GRADUATE RESEARCH ASSISTANT

Aug 2023 - May 2025

- Engineered hybrid AI algorithms integrating symbolic reasoning with data-driven learning.
- Emphasized robustness, assurance, and trustworthy deployment.

NSF Grant IIS-1652835 NSF

GRADUATE RESEARCH ASSISTANT

2021 - 2025

- Co-developed algorithms for scalable inference in graphical models.
- Supported publications recognized through best paper, spotlight, and oral presentations at NeurIPS and AAI.

AFOSR grant FA955021S0001 (CFDA 12.800) AFOSR

GRADUATE RESEARCH ASSISTANT

2021 - 2025

- Supported research on probabilistic logic and credibility modeling for multimodal learning.
- Advanced reliable learning methods designed to reduce label requirements.

Professional Recognition and Honors

Paper Awards and Selective Presentations

"A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models" selected as Spotlight: top 2% papers	NeurIPS 2024
Best Paper Award for "A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models"	TPM 2024
"Neural Network Approximators for Marginal MAP in Probabilistic Circuits" selected for oral presentation: top 3% papers	AAAI 2024

Reviewing Recognition

Recognized as a Silver Reviewer for ICML 2026	ICML 2026
Recognized as Top Reviewer (among the top 8% of reviewers) for NeurIPS 2024	NeurIPS 2024

Academic Honors and Fellowships

Certificate of Academic Excellence (PhD): Awarded for Outstanding Academic Performance in the Computer Science Graduate PhD Program	UTD
Graduate Teaching Certificate: Awarded by the Center for Teaching and Learning	UTD
Certificate of Academic Excellence (MS): Awarded for Outstanding Academic Performance in the Computer Science Graduate Masters Program	UTD
Awarded the Jonsson School Graduate Study Scholarship (2019)	UTD

Publications

Peer-Reviewed Conference Publications

1. **Shivvrat Arya**, Smita Ghosh, Bryan Maruyama, Venkatesh Srinivasan, “RELINK: Edge Activation for Closed Network Influence Maximization via Deep Reinforcement Learning”, *34th ACM International Conference on Information and Knowledge Management*, **CIKM**, 2025
2. Brij Malhotra, **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “Learning to Condition: A Neural Heuristic for Scalable MPE Inference”, *39th Annual Conference on Neural Information Processing Systems*, **NeurIPS**, 2025
3. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “SINE: Scalable MPE Inference for Probabilistic Graphical Models using Advanced Neural Embeddings”, *28th International Conference on Artificial Intelligence and Statistics*, **AISTATS**, 2025
4. Rohith Peddi, **Shivvrat Arya**, Bharath Challa, Likhitha Pallapothula, Akshay Vyas, Bhavya Gouripeddi, Qifan Zhang, Jikai Wang, Vasundhara Komaragiri, Eric Ragan, Nicholas Ruoizzi, Yu Xiang, Vibhav Gogate, “CaptainCook4D: A Dataset for Understanding Errors in Procedural Activities”, *The Thirty-eight Conference on Neural Information Processing Systems, Datasets and Benchmarks Track*, **NeurIPS: D&B**, 2024
5. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models”, *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, **NeurIPS Spotlight: top 2% papers**, 2024
6. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “Neural Network Approximators for Marginal MAP in Probabilistic Circuits”, *The 38th Annual AAAI Conference on Artificial Intelligence*, **AAAI Oral Presentation: top 3% papers**, 2024
7. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “Learning to Solve the Constrained Most Probable Explanation Task in Probabilistic Graphical Models”, *27th International Conference on Artificial Intelligence and Statistics*, **AISTATS**, 2024
8. **Shivvrat Arya**, Yu Xiang, Vibhav Gogate, “Deep Dependency Networks and Advanced Inference Schemes for Multi-Label Classification”, *27th International Conference on Artificial Intelligence and Statistics*, **AISTATS**, 2024
9. Vikas Chauhan, Aruna Tiwari, **Shivvrat Arya**, “Multi-Label classifier based on Kernel Random Vector Functional Link Network”, *International Joint Conference on Neural Networks*, **IJCNN**, 2020

Peer-Reviewed Journal Publications

1. Reza Shahriari, Amal Hashky, **Shivvrat Arya**, Tyler Audino, Eric D. Ragan, Vibhav Gogate, Jaime Ruiz, “Comparison of Text-Based Inputs for Human-in-the-Loop Feedback in Vision–Language Models”, *ACM Transactions on Interactive Intelligent Systems*, **TiiS**, 2026
2. Chiradeep Roy, Mahsan Nourani, **Shivvrat Arya**, Mahesh Shanbhag, Tahrima Rahman, Eric D. Ragan, Nicholas Ruoizzi, Vibhav Gogate, “Explainable Activity Recognition in Videos using Deep Learning and Tractable Probabilistic Models”, *ACM Transactions on Interactive Intelligent Systems*, **TiiS**, 2023

Peer-Reviewed Workshop Publications

1. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “Neural Network Approximators for Marginal MAP in Probabilistic Circuits”, *UAI Tractable Probabilistic Modeling*, **TPM**, 2024
2. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models”, *UAI Tractable Probabilistic Modeling*, **TPM: Best Paper Award**, 2024
3. Benjamin Rheault, **Shivvrat Arya**, Akshay Vyas, Jikai Wang, Rohith Peddi, Brett Benda, Vibhav Gogate, Nicholas Ruoizzi, Yu Xiang, Eric D Ragan, “Predictive Task Guidance with Artificial Intelligence in Augmented Reality”, *IEEE Conference on Virtual Reality and 3D User Interfaces*, **IEEE VR**, 2024
4. Rohith Peddi, **Shivvrat Arya**, Bharath Challa, Likhitha Pallapothula, Akshay Vyas, Qifan Zhang, Jikai Wang, Vasundhara Komaragiri, Nicholas Ruoizzi, Eric Ragan, Yu Xiang, Vibhav Gogate, “Put on your detective hat: What’s wrong in this video?”, *DMLR Data-centric Machine Learning Research*, **DMLR Workshop**, 2023

Preprints and Manuscripts Under Review

1. Yeqing Chen, Cong Qi, Hanzhang Fang, Feiyang Luan, Zhirong Zhang, **Shivvrat Arya**, Zhi Wei, “CoLa-VAE: Cell-Cell Communication-aware Variational Autoencoder with Dynamic Graph Laplacian Constraints”, *bioRxiv*, **Preprint**, 2026
2. Brij Malhotra, **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, “Learning to Guide Local Search for MPE Inference in Probabilistic Graphical Models”, *Preprint*, **Under Review**, 2026

Mentoring and Advising

Doctoral Committee

COMMITTEE MEMBER

- Mohith Oduru, Title: Spatially-Indexed Persistent Memory: A 3D-Indexed Latent Memory Module for Unsupervised Egocentric World Models (Qualifying Exam Committee member).
- Xiaoliang Wu, Title: Evaluating Reasoning-Intensive Retrieval through Information-Seeking and Causal Attribution in Stock Price Dynamics (Qualifying Exam Committee member).
- Hanzhang Fang, Title: Generative Spatial Deconvolution via Disentangled Latent Factors and Domain-Adaptive Decoder Transfer (Committee member).
- Yeqing Chen, Title: Cell-Cell Communication-aware Variational Autoencoder with Dynamic Graph Laplacian Constraints (Committee member).

Master's Thesis Committee

COMMITTEE MEMBER

- Kadir Altunel, Title: Edge Co-occurrence Regularization for Node Classification (Thesis Committee member).
- Sabdha Ambati, Title: Benchmarking AI Understanding of Adolescent Digital Language (Thesis Committee member).

Research Mentoring

Department of Computer Science, The University of Texas at Dallas

Richardson, TX

RESEARCH MENTOR

August 2022 - May 2025

- **Graduate Level:** Guided Ph.D. and Master's students in artificial intelligence, machine learning, and computer vision research, resulting in conference and journal publications, as well as successful thesis completions.
- **Undergraduate Level:** Mentored undergraduate students in artificial intelligence, machine learning, and computer vision research projects.
- **Responsibilities:** Provided technical guidance, reviewed research manuscripts, conducted regular progress meetings, and assisted in experimental design and implementation.

Classroom Teaching

Department of Computer Science, New Jersey Institute of Technology

Newark, NJ

COURSE INSTRUCTOR

August 2025 - May 2026

- **CS 370: Introduction to Artificial Intelligence - Honors:** Taught the honors undergraduate AI section as the sole instructor.
- **CS 785 ST: NeuroSymbolic AI:** Developed and taught a new Ph.D. seminar course expanding the department's AI curriculum; planned adaptation of course material for inclusion in the undergraduate CS curriculum.

Department of Computer Science, The University of Texas at Dallas

Richardson, TX

GUEST LECTURER

January 2024 - December 2024

- **Courses:** Machine Learning (Graduate Level); Artificial Intelligence (Graduate Level); Artificial Intelligence (Senior Level)
- **Responsibilities:** Delivered instructional content and moderated discussions on advanced Machine Learning concepts; Taught lectures and guided class discussions on advanced topics in Artificial Intelligence; Delivered lectures and facilitated class activities focused on core topics in Artificial Intelligence

Department of Computer Science, The University of Texas at Dallas

Richardson, TX

TEACHING ASSISTANT

August 2020 - August 2021

- **Courses:** Statistical Methods in AI and Machine Learning, Database Systems, Discrete Mathematics for Computing II
- **Responsibilities:** Graded assignments; Held office hours; Conducted project demonstrations; Developed supplementary materials and homework solutions

Summer School Programs

India

HIGH SCHOOL TEACHER

Summer 2015, 2016, 2017

- **Earlier Teaching Experience:** Taught computer science and mathematics in summer school programs in India (2015, 2016, 2017).

Certificates

Center for Teaching and Learning, The University of Texas at Dallas

Richardson, TX

GRADUATE TEACHING CERTIFICATE RECIPIENT

2025

- Completed formal training in instructional pedagogy through the UTD Center for Teaching and Learning.
- Fulfilled four core components: Epigeum coursework, faculty teaching observation, reflective essays, and semester-long teaching experience.
- Certificate awarded upon demonstrating teaching effectiveness and commitment to professional development in higher education.

Talks and Presentations

Invited Talks

NJIT INFO Seminar	<i>Newark, NJ, USA</i>
KNOWING WHAT COMES NEXT AND WHAT WENT WRONG: HUMAN-CENTERED AI FOR AR PROCEDURAL GUIDANCE	<i>Mar 2026</i>
Santa Clara University	<i>Santa Clara, CA, USA</i>
NEURAL SOLVERS FOR FAST, ACCURATE PROBABILISTIC INFERENCE	<i>Jan 2026</i>

Award Talks and Presentations

The 7th Workshop on Tractable Probabilistic Modeling (TPM)	<i>Barcelona, Spain</i>
A NEURAL NETWORK APPROACH FOR EFFICIENTLY ANSWERING MOST PROBABLE EXPLANATION QUERIES IN PROBABILISTIC MODELS (BEST PAPER AWARD)	<i>May 2024</i>
The 38th Annual AAAI Conference on Artificial Intelligence	<i>Vancouver, Canada</i>
NEURAL NETWORK APPROXIMATORS FOR MARGINAL MAP IN PROBABILISTIC CIRCUITS (ORAL PRESENTATION)	<i>Mar 2024</i>

Professional Service

I regularly review papers for the following venues:

Journal Reviewer

Artificial Intelligence (AIJ)
Transactions on Machine Learning Research (TMLR)
IEEE Robotics and Automation Letters (RA-L)
Data Mining and Knowledge Discovery

Conference Reviewer

Neural Information Processing Systems (NeurIPS: Recognized as a Top Reviewer in 2024)
International Conference on Machine Learning (ICML: Recognized as a Silver Reviewer in 2026)
International Conference on Learning Representations (ICLR)
Conference on Uncertainty in Artificial Intelligence (UAI)
International Conference on Artificial Intelligence and Statistics (AISTATS)

Organizational Responsibilities

Completed at New Jersey Institute of Technology, Department of Computer Science

Member	Undergraduate Committee (Fall 2025-Present)
Member	AI Curriculum Committee (Fall 2025-Present)
Member	Research Committee (Fall 2025-Present)
Member	Grade Appeal Committee (Fall 2025-Present)
Member	AI Innovations Committee (Fall 2025-Present)

Completed at The University of Texas at Dallas

Associate Officer Internal Projects	Research Club
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Completed at Indian Institute of Information Technology Vadodara, India

Head	Sports Committee
Core Member	Organizing Team – Krieva 2016, Ventura 2016, Krieva 2017, Cerebro 2018, Ventura 2018
Core Member	Pensive (Literary Society)
Member	Hostel Executive Committee
Class Representative	Vidya Sagar School

Technical Skills

Programming Languages:	Python, R, SQL, C/C++, Java, Bash, Cython
Machine Learning:	PyTorch, TensorFlow, scikit-learn, JAX, Keras, XGBoost
Deep Learning:	CNNs, Transformers, Vision-Language Models, Representation Learning, Generative Models
Neuro-Symbolic AI:	Probabilistic Reasoning, Knowledge Graphs, Logic Programming, Semantic Parsing
Computer Vision:	OpenCV, PIL, torchvision, Detection, Segmentation, Multimodal Learning
Data Science:	NumPy, Pandas, SciPy, Matplotlib, Seaborn, Jupyter
ML Infrastructure:	Docker, Kubernetes, Git, GitHub, GitHub Actions, MLflow, Weights & Biases
Cloud and HPC:	AWS, Google Cloud Platform, Azure, SLURM, CUDA

Academic Projects

Research Software

- **NeuPI: A Library for Neural Probabilistic Inference** **Research Software, 2026**
Developed a PyTorch-based library for neural probabilistic inference, supporting self-supervised neural surrogates, ITSELF test-time adaptation, and MPE, Marginal MAP, and Constrained MPE tasks across probabilistic graphical models, probabilistic circuits, and neural autoregressive models.

Completed at The University of Texas at Dallas

- **Parameter and Structure Learning Algorithms for Bayesian Networks** **Statistical Methods in AI and ML, Spring 2020**
Implemented several structure learning algorithms for Bayesian Networks, including FOD-Learn (fully observed data, known structure), POD-Learn (partially observed data, known structure), and Mixture-Random-Bayes (fully observed data, unknown structure). Conducted a comparative analysis of these algorithms based on different data and structure conditions.
- **Sampling-based Variable Elimination and Conditioning** **Statistical Methods in AI and ML, Spring 2020**
Implemented Sampling-based Variable Elimination and Conditioning algorithm for performing inference on probabilistic graphical models.
- **Learning Algorithms for Bayesian Networks** **Machine Learning, Fall 2019**
Implemented four algorithms: Independent Bayesian Networks, Tree Bayesian Networks (using Chow-Liu), Mixtures of Tree Bayesian Networks using EM, and Mixtures of Tree Bayesian Networks using Random Forests.
- **Non-Iterative Neural Networks** **Machine Learning, Fall 2019**
Implemented two non-iterative neural network models for classification and regression tasks across multiple datasets.
- **Collaborative-Filtering** **Machine Learning, Fall 2019**
Implemented collaborative filtering algorithms using a subset of the Netflix Prize movie ratings data. Evaluated model performance using Mean Absolute Error and Root Mean Squared Error metrics.
- **DART Database System** **Database Design, Fall 2019**
Designed and implemented the DART system's complete database, from EER diagram creation to relational schema design, SQL-based database construction, and query/view generation.

Completed at Indian Institute of Information Technology Vadodara, India

- **Compressed Sensing** **Computer Vision, Fall 2018**
Developed and evaluated an algorithm for multi-view tracking and 3D voxel reconstruction using 2D images.
- **Kakuro Puzzle Solver** **Artificial Intelligence, Spring 2018**
Designed and implemented a bot to solve Kakuro puzzles by applying derived rules to handle diverse puzzle instances.
- **Autoencoder for Anomaly Detection** **Deep Learning, Spring 2018**
Developed an autoencoder-based anomaly detection method, successfully tested on a credit card fraud dataset.
- **LEARN: A Programming Language** **Compiler Design, Spring 2018**
Designed and implemented a beginner-friendly programming language LEARN using yacc and lex, aiming to ease the learning curve for new programmers.
- **Fatal Disease Detector Using Twitter Data** **IIITV Hackathon, Fall 2018**
Implemented a k-means clustering algorithm to detect disease spread patterns using Twitter data.

- **SoT (Security of Things)** **Cryptography, Fall 2017**
Developed an Android application to monitor real-time environmental temperature changes with AES encryption for secure data transmission.
- **Hatsphere: E-commerce Platform** **Software Development, Fall 2017**
Developed a platform to enable traditional craftsmen to sell their products online and reach a broader market.
- **Cocktail Party Effect Algorithm** **Speech Science and Technology, Fall 2017**
Implemented an algorithm to separate human voice from noise in audio recordings.
- **Movie Recommendation System** **Database Design, Spring 2016**
Developed a user interface and managed a database system for a movie recommendation engine, utilizing PostgreSQL and C for backend management.

Certifications

Coursera Course Certificates

- **Mathematics for Machine Learning: Linear Algebra** – Present License E5PBMECK8B4M Apr 2019
- **What is Data Science?** – Present License 5WS64BF2G2SY Feb 2019
- **Introduction to Programming with MATLAB** – Present License VGPWCM8WH73K Feb 2019
- **Deep Learning Specialization** – Present License J9V32CC6VTB5 Nov 2018
- **Sequence Models** – Present License 5E9VFH59THG4 Nov 2018
- **Convolutional Neural Networks** – Present License G888N3WXPXLN Sep 2018
- **Python Programming Essentials** – Present License TZXYXTF796L Jun 2018
- **Structuring Machine Learning Projects** – Present License JJNYAQTTFUPR May 2018
- **Neural Networks and Deep Learning** – Present License JSAR6KKVC5Y7 Apr 2018
- **Improving Deep Neural Networks** – Present License MRVNDFNWUHPQ Apr 2018
- **Machine Learning** – Present License FPWNJ39A5LWQ Mar 2018
- **Python Data Structures** – Present License DS5N3NM69PQ6 Mar 2018
- **Programming for Everybody** – Present License XS6H2XUBJ66U Jan 2018

Udemy Course Certificates

- The Top 5 Machine Learning Libraries in Python Jan 2019
- MATLAB for scientists: a beginner's course Jan 2019

Other Certifications and Professional Development

- Graduate Teaching Certificate Center for Teaching and Learning, The University of Texas at Dallas